

INVENTORY TRACKING SYSTEM

Java Console Application — OOP Project Report

Language: Java | File: InventorySystem.java | JDK 8+ | No External Libraries

Project Overview

A console-based inventory management program built in Java. It demonstrates core **OOP concepts** — encapsulation, class design, constructors with the **this** keyword, getters/setters, and loop-based iteration — through a real-world product management scenario.

Features

No.	Feature
01	Add products with name, stock quantity, and price
02	Display full inventory in a formatted numbered table
03	Auto-detect low-stock items and print alerts
04	Update stock quantity for any product by name
05	Calculate total stock count across all products

OOP Concepts Used

Concept	Where Applied
Encapsulation	All Product fields are private — accessed only via methods
this keyword	Constructor uses this to resolve parameter name conflicts
Getters	getName() getStockQuantity() getPrice()
Setters	setStockQuantity() setPrice()
toString() override	Custom formatted string output for each Product object
Index-based loop	displayInventory() — prints numbered product list
Enhanced for-loop	checkLowStock() updateStock() getTotalStock()

Project Structure

```
InventorySystem.java
|
+-- class Product
|   +-- Fields      : name, stockQuantity, price    (all private)
```

```
|    +-- Constructor : Product(name, stockQuantity, price)
|    +-- Getters    : getName(), getStockQuantity(), getPrice()
|    +-- Setters    : setStockQuantity(int), setPrice(double)
|    +-- toString() : formatted output string
|
+-- class InventoryManager
|    +-- Fields      : List<Product> products, int lowStockThreshold
|    +-- Methods     : addProduct(Product)
|                     displayInventory()
|                     checkLowStock()
|                     updateStock(String, int)
|                     getTotalStock()
|
+-- public class InventorySystem
    +-- main() <-- program entry point
```

■ How to Run

Requirement: Java JDK 8 or above must be installed.

```
# Step 1 – Compile
javac InventorySystem.java
```

```
# Step 2 – Run
java InventorySystem
```

■ Sample Output (Actual Terminal Screenshot)

```
PS C:\Users\MOHAMMED AADIL> cd "c:\Users\MOHAMMED AADIL\Downloads\" ; if ($?) { javac InventorySystem.java } ; if ($?) { java InventorySystem }

=====
INVENTORY STATUS REPORT
=====
1. Wireless Mouse      | Stock: 45 | Price: Rs. 799.00
2. Mechanical Keyboard | Stock: 8  | Price: Rs. 2499.00
3. USB-C Hub           | Stock: 3  | Price: Rs. 1299.00
4. Laptop Stand        | Stock: 22 | Price: Rs. 1099.00
5. Webcam HD           | Stock: 7  | Price: Rs. 3499.00
6. Desk Lamp           | Stock: 0  | Price: Rs. 599.00
7. Monitor Arm         | Stock: 15 | Price: Rs. 1899.00
8. HDMI Cable          | Stock: 60 | Price: Rs. 299.00
=====
Total Products: 8
=====

-----
LOW STOCK ALERTS (Limit: 10)
-----
? Mechanical Keyboard ? only 8 units left!
? USB-C Hub ? only 3 units left!
? Webcam HD ? only 7 units left!
? Desk Lamp ? only 0 units left!
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Restocking USB-C Hub...
Updated: USB-C Hub ? new stock = 25
Restocking Desk Lamp...
Updated: Desk Lamp ? new stock = 12

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```

■ Customization Guide

Task	Code
Change low stock limit	InventoryManager manager = new InventoryManager(15);
Add a new product	manager.addProduct(new Product("Name", qty, price));
Update product stock	manager.updateStock("Product Name", newQty);